

CRABS AND OTHER CRUSTACEANS



THERE ARE SOME 30,000 SPECIES in the crustacean class, including crabs, barnacles, copepods, krill, lobsters, prawns, shrimps, and woodlice. Crabs and other crustaceans share characteristics including two pairs of antennae, mandibles, and a shell. In size they can range from microscopic freshwater fleas to giant Japanese spider crabs, which have a claw-to-claw span of 3.6m (11 ft). Most crustaceans live in the sea, others prefer fresh water, and a few, such as the woodlouse, live on land. Some are parasitic – they live on or in other animals.

Teeth



Defence

Apart from the protection they get from their shells, crabs can also defend themselves using their enlarged claws, or pincers, which often have a sharp serrated (toothed) edge. Some crabs, such as the decorator crabs, camouflage themselves by fixing seaweed to their shells, and blending into their surroundings. However, even with these methods of defence, crabs are still eaten by octopuses, fish such as bass, shore birds, and mammals.

Claw of Japanese spider crab

Seaweed and shells attached to crab

Decorator crab



Features of a crab

Crabs belong to the order Decapoda (10-legged). They have four pairs of jointed walking or swimming legs, plus an extra modified pair, called pincers. They have two pairs of antennae, gills, and a segmented, calcareous shell, or carapace. As they grow, crabs shed the carapace and grow a new one.

Feeding

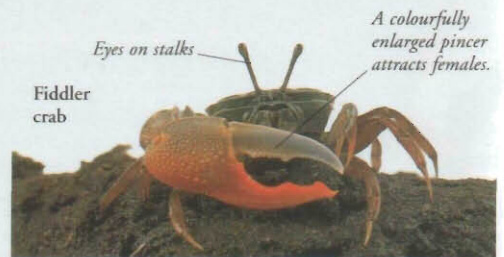
Crabs, such as the hermit crab, are generally omnivorous, and can be either predators or scavengers. Hermit crabs catch their prey in their pincers. The crab then uses the pincers like a fork, to pass the food to its mandibles (specialized jaws). The crab chews its food, then uses two pairs of adapted limbs to push the food further into the mouth.



Fish Hermit crab

Breeding

After the male has fertilized the female's eggs, she may carry them in a brood pouch until ready to hatch, or release them into the sea. Some crustaceans hatch as tiny adults, but crabs go through a larval stage, and spend their early life as plankton.



Fiddler crab

Eyes on stalks

A colourfully enlarged pincer attracts females.

How a crab moves

Crabs walk slowly forwards or backwards on their jointed legs when exploring their surroundings under water, but scuttle sideways across the ground when threatened. Some crabs, such as velvet crabs, are able to swim because their hind legs have been adapted into flattened paddles, also known as swimmerets.



Shore crab turning.

Missing limb

Small flap-like abdomen

Fully turned crab can scuttle away.

Barnacles

Young barnacles float in the sea until they find any stable surface, such as a rocky shore. They then attach themselves to a rock, and secrete a calcium-based substance that forms protective plates around them. They leave an opening for feeding, which closes when the animal is exposed to the air at low tide. Barnacles are hermaphrodites – that is, both sexes exist in one individual.

Goose barnacles attach themselves to driftwood with long stalks.



Goose barnacles

Feeding

Most barnacles feed themselves using their feet! When covered by water, these fine, feathery, curly limbs protrude from the barnacle's opening. The feet trap food as it floats past, filter it from the seawater, and transfer it into the barnacle's mouth. Goose barnacles do this while floating in the sea.

Woodlice

Woodlice are the only true land-living crustaceans. A woodlouse's "shell" consists of flat, waterproof plates that protect its back. Woodlice can dry out, so some species, such as the pill bug, have developed the ability to roll up and reduce water loss.

Waterproof plates

